helenjin@seas.upenn.edu • https://helenjin.github.io

Helen Jin

CURRICULUM VITAE - PREPARATION DATE: MAY 2026

RESEARCH INTERESTS

Keywords: Trustworthy Machine Learning, Explainability, Interpretability, Natural Language Understanding, Human–AI Interaction, Cognition, Education, Scientific Discovery

We live in a nascent, revolutionary era. How will we integrate Artificial Intelligence (AI) into our human society? Broadly speaking, I am interested in problems related to "AI for Humanity." I aim to develop AI systems that are safe, reliable, and genuinely beneficial in high-stakes, real-world settings. *My research is thus guided by a central question:* **How can we establish and sustain principled trust between humans and AI systems?** To address this, I develop methods to evaluate to what extent AI systems are reliable, aligned with domain expertise, and *correct for the right reasons*. I am also interested in using AI as a technical lens to reveal, formalize, and better understand (human) cognition and reasoning. Alongside my research, education and service are deeply important to me; hence, this commitment is reflected throughout my life, as showcased in this curriculum vitae.

EDUCATION

University of Pennsylvania, School of Engineering and Applied Science, Philadelphia, PA — Sep 2020 – Present

Ph.D. Candidate in Computer and Information Science (CIS)

*Received **MSE** in CIS along the way*

- **Advisor:** Prof. Eric Wong
- Served on the Graduate and Professional Student Assembly (GAPSA) as part of the Executive Board, actively advocating for and working toward improving graduate student life across all twelve schools of Penn
- Perry World House Graduate Associate | AWS-AI ASSET Fellow | CETLI Teaching Certificate

Columbia University, Columbia College, New York, NY — Sep 2016 - May 2020

B.A. – Double Major in Mathematics and Computer Science*, Concentration in East Asian Studies

Computer Science Track: Intelligent Systems — *Minimal overlap in courses

- **Advisor:** Prof. Tian Zheng (worked on weakly supervised learning problems)
- **Relevant Coursework**: Data Structures in Java, Advanced Programming in C/C++, Discrete Mathematics, Fundamentals of Computer Systems, Computer Science Theory, Artificial Intelligence, Machine Learning, Natural Language Processing, Computer Vision, Computation and the Brain, Advanced Spoken Language Processing, Databases, Linear Algebra, Modern Algebra I + II, Modern Analysis I + II, Topology, Calculus-Based Statistics, Probability Theory
- **Honors/Awards**: Dean's List

Girls Who Invest, The Wharton School, University of Pennsylvania, Philadelphia, PA — May 2018 – Aug 2018

Summer Intensive Program Scholar

- Completed highly selective, rigorous 4-week training program focused on core investment concepts and skills taught by leading business school professors and investment professionals; upon completion, earned a paid 6-week internship in asset management
- Coursework and case studies included accounting, valuation, financial modeling, asset allocation and presentation skills

Stuyvesant High School, New York, NY — Sep 2012 – Jun 2016

Advanced Regents Diploma

- **Activities**: Girls' Varsity Swimming, Lifeguard, Stuyvesant Spectator, Math Team, CS Dojo and Writing Center Tutor
- **Honors/Awards**: AP National Scholar; National Merit Finalist; ARISTA National Honor Society; National Latin Society; National Junior Classical League; National Latin Exam Awards: Maxima Cum Laude; Award of Distinction in Mathematics

PUBLICATIONS (Google Scholar)

[17] “AFFIX: Automated Feature Finder for Interpretable Explanations.” *Ongoing.*

[16] “Stability Guarantees for Scientific Claim Verification.” *Ongoing.*

[15] “Oral Assessments: Design and Evaluation of Assessments for Graduate Level Computer Science Assignments.” **Helen Jin,** Arnab Sircar, Xiaoshen Ma, Surbhi Goel, Eric Wong. *Ongoing.* ***Most Active.***

[14] “Diagnosing How Scientific Reasoning Fails: A Benchmark for Qualifier and Rebuttal Errors.” **Helen Jin,** Eric Wong. *In Submission.* ***Most Active.***

[13] “Reliability, Faithfulness, and the Limits of Post-hoc Explanations of Opaque Scientific Models.” Nick Oh, **Helen Jin.** *In Submission.*

[12] “Co-TA: Streamlining Automated Grading with Generative Rubrics and Distribution Simulation.” **Helen Jin,** Albert Chen. *In Submission.*

[11] “T-FIX: Text-Based Explanations with Features Interpretable to eXperts.” Shreya Havaldar*, Weiqiu You*, Chaehyeon Kim* Anton Xue*, **Helen Jin***, Gary Weissman, Rajat Deo, Sameed Khatana, Helen Qu, Marco Gatti, Daniel A. Hashimoto, Amin Madani, Masao Sako, Bhuvnesh Jain, Lyle Ungar, Eric Wong. 2025. *In Submission.* [arXiv]

[10] “Probabilistic Soundness Guarantees in LLM Reasoning Chains.” Weiqiu You, Anton Xue, Shreya Havaldar, Delip Rao, **Helen Jin**, Chris Callison-Burch, Eric Wong. In *EMNLP 2025.* [arXiv] [Paper]

[9] “Probabilistic Stability Guarantees for Feature Attributions.” **Helen Jin***, Anton Xue*, Weiqiu You, Surbhi Goel, Eric Wong. 2025. In *NeuRIPS 2025.* [arXiv] [Paper] [Poster] [Blog]

[8] “Adaptive Evaluations: Profiling model capabilities with evaluator agents.” Davis Brown, Prithvi Balehannina, **Helen Jin**, Shreya Havaldar, Hamed Hassani, Eric Wong. In *EMNLP 2025*. [arXiv] [Paper]

[7] “FIX: A Benchmark for Features Interpretable to eXperts.” **Helen Jin***, Shreya Havaldar*, Chaehyeon Kim*, Anton Xue*, Weiqiu You*, Helen Qu, Marco Gatti, Daniel A Hashimoto, Bhuvnesh Jain, Amin Madani, Masao Sako, Lyle Ungar, Eric Wong. In *DMLR 2025*. [arXiv] [Paper] [Blog] [Website]

[6] “Certifiably Robust Evaluation of Feature Attributions via Boolean Influences.” **Helen Jin**, Weiqiu You, Eric Wong. 2024. [Preprint]

[5] “Linguistic Properties of Truthful Text Generation.” Bruce W. Lee, Benedict Florance Arockiaraj, **Helen Jin**. In *TrustNLP: Third Workshop on Trustworthy Natural Language Processing at ACL 2023.* [arXiv]

[4] “Automatically Generated Summaries of Video Lectures Enhance Students' Learning Experience.” Hannah Gonzalez*, Jiening Li*, **Helen Jin***, Jiaxuan Ren*, Hongyu Zhang*, Ayotomiwa Akinyele*, Adrian Wang, Eleni Miltsakaki, Ryan Baker, Chris Callison-Burch. In *The 18th Workshop on Innovative Use of NLP for Building Educational Applications (BEA) at ACL 2023.* [Paper]

[3] “Generic Temporal Reasoning with Differential Analysis and Explanation.” Yu Feng, Ben Zhou, Haoyu Wang, **Helen Jin**, Dan Roth. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL).* 2023. [arXiv] [Paper]

[2] “Large-scale, image-based tree species mapping in a tropical forest using artificial perceptual learning.” Tang, C, Uriarte, M, **Jin, H**, Morton, DC, Zheng, T. *Methods Ecol Evol*. 2021; 12: 608– 618. [Paper]

[1] "Artificial Perceptual Learning: Image Categorization with Weak Supervision." Chengliang Tang, Maria Uriarte, **Helen Jin**, Douglas Morton, Tian Zheng. 2021. [arXiv]

*Indicates equal contribution

FEATURED MEDIA

- "We are losing the point of education" (2026)
  - Op-ed appearing in The Daily Pennsylvanian on education in the age of AI and how courses like CIS 5200 (Machine Learning) are exploring new assessment strategies, where I designed and led the scalable adoption of oral assessments in this course
- "The ASSET Center Advances Trustworthy AI with Support from AWS" (2025)
- "Evaluating Large Language Models for Cyberbullying Behavior" (2025)
  - News article appearing in Penn Today about our cyberbullying research
- "AI Meets Medicine: Penn Secures ARPA-H Funding for AI in Breast Cancer Care and More" (2024)
- "Penn Engineering's ASSET Center: An Interdisciplinary Initiative Building Safe, Explainable and Trustworthy AI Systems" (2023)
- "2022-2023 GAPSA Leadership Major Accomplishments and Historic Wins Letter" (2023)

WORK EXPERIENCE

BrachioLab, University of Pennsylvania, Philadelphia, PA — Sep 2023 – Present

Research Assistant
- PI: Prof. Eric Wong
- Working on problems in Explainable Artificial Intelligence (XAI), specifically the evaluation of explanations

Amazon (AWS), New York, NY — May 2025 – Aug 2025

Applied Scientist Intern
- Manager: Yi Fan
- Mentors: Ecenaz Erdemir, Bugra Can, Bhavna Soman
- Security Analytics and AI Research (SAAR) team
  - Agentic AI for Security Domain, Runtime Threat Learning

Cognitive Computation Group, University of Pennsylvania, Philadelphia, PA — Sep 2020 – Sep 2023

Research Assistant
- PI: Prof. Dan Roth
- Worked on Natural Language Understanding (NLU) problems, including Complex Event Annotation Project and The State of the Art of Semantic Role Labeling (SRL) Systems

University of New York-Herbert H. Lehman College, Virtual — Jun 2020 – Aug 2020

PTS3 Data Science Mentor-Instructor
- Pathways to Student STEM Success Program (PTS3)
- Supervised the capstone projects of four undergraduate students who previously had little/no prior experience with data science
- Mentored and taught students relevant technology for their projects such as Python, R, Google Colab

Department of Statistics, Columbia University, New York, NY — Jun 2019 – Aug 2020

Undergraduate Researcher
- PI: Prof. Tian Zheng
- Worked on a research project that seeks to develop an artificial perceptual learning framework that can aid semi-supervised learning problems, specifically with a focus on image classification on large images of tree canopies with a small labeled tree species data set
- Used various statistical and deep learning methods primarily in Python, including TensorFlow and Keras packages

LionBase, LLC, New York, NY — Dec 2018 – Sep 2019

Client-Facing Team Member

- LionBase is a new student-led data science and product development group at Columbia University
- Worked in a team of six to solve real-world industry problems related to data analytics, ML, NLP, and statistical analysis
- Spring 2019: Recruitment Platform for Early Stage Data Science Talent – my team collaborated with an executive search firm to design, develop, and implement from scratch a platform that matches data science talent and company based on various technical and non-technical tests.

Data Science Institute, Columbia University, New York, NY — Mar 2019 – May 2019

Data for Good (DFG) Scholar

- Worked in a team in collaboration with Okimo, a startup using eye tracking technology to help identify and aid individuals' reading skills especially targeting younger age groups in low resource communities in developing countries such as Paraguay

Grantham, Mayo, van Otterloo & Co. LLC (GMO), Boston, MA — Jul 2018 – Aug 2018

Quantitative Equity Analyst Intern

- Using MATLAB and SQL, created a working tool to measure the relationship between institutional investor concentration and downside risk in equities, and modeled risk in global equity markets

Computational and Systems Biology, Memorial Sloan Kettering Cancer Center, New York, NY — May 2017 – Aug 2018

Undergraduate Researcher

- PI: Prof. Dana Pe'er
- Developed computational methods and packages to analyze single cell RNA-seq data using Python and R

TEACHING EXPERIENCE

"Thus it is proper to the role of the scientist that he not merely find the truth and communicate to his fellows, but that he teach, that he try to bring the most honest and most intelligible account of new knowledge to all who will try to learn."

– J. Robert Oppenheimer

I have always had a strong love and mission for teaching and mentoring. Throughout the years, I have naturally found myself TA-ing over 10+ courses (often taking on multiple courses in the concurrent semester) and taking on instructor roles simply because I find teaching incredibly rewarding and I believe it is essential to give back the knowledge I have learned as a scholar to the next generation. I hope to continue to foster this passion and commitment to teaching as I progress in my career, both in formal and informal capacities wherever I can. Distilled at its core, teaching is about helping others augment and realize their potential so that they may enact positive change to the world and inspire in their own ways. Is there anything better than this noble honor?

University of Pennsylvania, Philadelphia, PA

- **Teaching Assistant**, CIS 5200: Machine Learning [Spring 2026]
  - I led the design and deployment of a novel oral assessment (OA) framework in a graduate-level ML course (~200 students), which was the university's first large-scale implementation replacing traditional grading of written homework with oral assessment. The OA framework leverages LLMs to convert written solutions into structured, specific oral rubrics, enabling scalable evaluation and prioritizing authentic student understanding in the presence of generative AI tools.
- **Teaching Assistant**, EAS 2030: Engineering Ethics [Fall 2025]
  - This is a required ethics course that all undergraduate Engineering students must take (~200 students).
- **Teaching Assistant**, CIS 3333: Mathematics of Machine Learning [Fall 2024]
- **Teaching Assistant**, CIS 4210/5210: Artificial Intelligence [Fall 2021]

    - This is the largest course the university offers (~500 students). I managed 2 sections of the course simultaneously and single-handedly assigned homework pairs.
- **Teaching Assistant**, Girls Who Invest Summer Intensive Program @ The Wharton School [Summer 2023]

University of New York-Herbert H. Lehman College, New York, NY

- **Data Science Mentor-Instructor**, Pathways to Student STEM Success Program (PTS3) [Summer 2020]
    - Served as a Data Science Mentor-Instructor, leading in-classroom instruction and providing one-on-one mentoring to undergraduate students who previously had little to no background in data science. Guided students through their first data science projects, building foundational and quantitative skills from the ground up and delivering supplementary lessons in Python and R adapted to each student's skill level.

Columbia University, New York, NY

- **Teaching Assistant,** Natural Language Processing [Fall 2019, Spring 2020, Summer 2020]
- **Teaching Assistant,** Linear Algebra [Spring 2019]
- **Teaching Assistant,** Calculus IV [Spring 2018, Summer 2019]
- **Teaching Assistant,** Calculus III [Fall 2018, Summer 2019]
- **Teaching Assistant,** Calculus I [Fall 2017]
- **Teaching Assistant,** Calculus-Based Intro to Statistics [Fall 2017]

Other

- **Private Tutor** for Natural Language Processing course at Columbia University [Spring 2021]
- **Private Tutor** for elementary school children, with specific exposure to computer science concepts and coding [Summer 2020]
- **In-Person Private Tutor** for Mathematics and Chemistry subjects, high school level [Feb 2017 – Jun 2017]
- **One-to-One Tutoring Volunteer** - Volunteered weekly to tutor and mentor 6-12 year olds in Harlem community in NYC [2016-2019]

SERVICE ACTIVITIES

- **Reviewer** for: ACL, China National Conference on Computational Linguistics (CCL) , NAACL, NeuRIPS, DMLR
- **Organizer** of: Computational Linguistics (CLunch) seminar at the University of Pennsylvania (Spring 2023)
- **Conference Volunteer** for: NAACL (2023), NeuRIPS (2025)
- **CIS Department PhD Mentorship Program Mentor + Event Coordinator Volunteer** (AY 2022-2023, AY 2023-2024, AY 2024-2025)

University of Pennsylvania, Philadelphia, PA

Served as a **Representative** for the following:

- **University Council** [AY 2021-2022, AY 2022-2023]
- **University Committee on Honorary Degrees** [AY 2021-2022]
- **Student Advisory Group for the Environment (SAGE)** [AY 2021-2022]
- **University Committee of Academic Affairs** [AY 2022-2023]
- **Ivy+ Summit Leadership Conference** [AY 2021-2022 hosted by University of Pennsylvania, AY 2022-2023 hosted by Columbia University]

SELECTED LEADERSHIP EXPERIENCE

Graduate and Professional Student Assembly (GAPSA), University of Pennsylvania, Philadelphia, PA

https://gapsa.upenn.edu

Director of Cultural Programming May 2024 – May 2025

Vice President of Operations May 2022 – May 2023

- Directly oversaw and managed the Operations Division officers

● Solely managed a ~$100K budget
● Served as primary liaison to student governments of each of the graduate and professional schools
● Worked on Pilot GAPSA Meal Swipe Program that passed in the General Assembly– $50,000 of the graduate student government budget was used to provide free meal swipes to students facing food insecurity [Resolution]
● [2022-2023 GAPSA Leadership Major Accomplishments and Historic Wins Letter]

Director of Logistics May 2021 – May 2022

Girls Who Code (GWC) at Columbia University, New York, NY Feb 2017 – Mar 2020
https://outreach.engineering.columbia.edu/content/girls-who-code

President
● Oversaw high school outreach and recruitment for program that holds weekly CS classes for ~50 high school girls each semester
● Planned on-site visits, created fundraisers, and reached out to companies and local businesses for sponsorship
● Worked with other Executive board members to better improve program
● Managed ~30 people on the Managing Board (Finance, Programming, Publicity, HS Recruitment)
● Previously served as **Vice President of External Affairs**, and before that, **Sponsors and Finance Team**

One-to-One Tutoring at Columbia University, New York, NY Oct 2016 – Sep 2019
https://one2onetutoring.wordpress.com/

Treasury Coordinator, Head Coordinator
● Allocated and managed annual $2,000 budget to various on-campus events from weekly meetings to fundraisers
● Supervised and oversaw volunteers during tutoring sessions each semester to ensure attendance
● Worked with other coordinators and Community Impact staff to discuss issues and implement solutions to improve program
● Volunteered weekly to tutor and mentor 6-12 year old student throughout academic year

Smart Woman Securities (SWS), Columbia Chapter, New York, NY Sep 2017 – May 2018

Senior Research Analyst
● Led a team of SWS Analysts researching the Consumer industry
● Conducted background research, analyzed and formulated investment opinions on companies to make real-life investment recommendations

SELECTED OTHER PROJECTS

Semantic Role Labeling (2022)
● Research on predicate and argument identification and classification

Complex Events (2022)
● Research on complex event extraction and knowledge graphs

Persistent Homology (2019)
● Created a visualization tool for the persistent homology in Python for my Math seminar on Elementary Applied Topology

SCAnalysis (2018)
● Built a Python package and Jupyter notebook to analyze single cell RNA-seq data that incorporates MAGIC and Wishbone technologies, with additional features (including Palantir). Wishbone is an algorithm to align single cells from differentiation systems with bifurcating branches. MAGIC (Markov-Affinity Based Graph Imputation of Cells) is an interactive tool to impute missing values in single-cell data and restore the structure of the data.

Other (more recent projects):
- "Cyberbullying Evaluation for LLM's" (2025)

- “Runtime Threat Learning and Analysis System with Agent Orchestration” (2025)
- “Safe and Explainable AI-Enabled Decision Making For Personalized Clinical Decision Support” (2025)

HONORS & AWARDS

- **AWS-AI ASSET Fellow** (2025)
    - Gift from AWS AI to Penn Engineering's ASSET Center for Trustworthy AI [Article]
- **Penn Wharton Innovation Fund Validation Award** (2023)
- **Dean’s List**, Columbia University (2016 - 2020)
- **Rewriting the Code (RTC) Fellow** (2017- present)
- **Science Technology Engineering Program (STEP) Award**, Columbia University (2017)

SKILLS & INTERESTS

Technologies: Java, Python, MATLAB, SQL, R, C/C++, Git, Jupyter Notebook, LaTeX , HTML/CSS/JavaScript, Coq

Languages: English, Korean, Mandarin Chinese, some Spanish and Japanese, learning French

Personal Interests: Psychology, Philosophy, Music, Visual Arts, Swimming, Yoga, Pilates, Traveling, Hiking, Squash, Tennis

REFERENCES

Can provide references upon request.